BASKETBALL MANAGER

Project Overview & Technical Reference

Author	Arya Chakraborty
Engine	Godot 4.6.2 (GDScript)
Platform	PC (macOS / Windows / Linux)
Database	SQLite via godot-sqlite plugin (AssetLib)
Started	April 2026
Status	In Development — Foundation & Data Layer Complete

1. Vision & Concept

Basketball Manager is a deep, simulation-based basketball management game inspired by Football Manager and Out of the Park Baseball. The player takes the role of a head coach, managing every aspect of a basketball program across multiple interconnected leagues.

Unlike existing basketball management games, this aims for Football Manager depth: a rich simulation engine that evaluates 53 player attributes per action, a living world where AI teams make intelligent decisions, and a career mode where the coach's reputation grows organically from G League gyms to NBA arenas.

Core Design Pillars

Pillar	Detail
Simulation depth	Every match outcome traces back to 53 individual player attributes across 7 categories
A living world	AI teams trade, sign free agents, draft players, and develop rosters independently
Multi-league ecosystem	NBA, G League, college, and overseas leagues — all connected and config-driven
Career progression	Coach reputation gates job opportunities; build from obscurity to dynasty
No 3D match engine	Text-based simulation with detailed play-by-play commentary

Inbox/news system	FM/2K-style message inbox — every significant event lands as a readable message

2. Leagues & Scope

The player selects which leagues to include when starting a new game. Each league is config-driven via JSON and can be toggled independently.

League	Details
NBA (30 teams)	Fully playable. Full salary cap system, luxury tax, max contracts, two-way deals, 15-man roster, G League affiliates.
G League (30 teams)	Full team entities with own identity, standings, and coaching career path. 1-year flat contracts. Inherits NBA affiliate owner personality.
College Basketball	Recruiting pipeline into the draft. Fictional programs. 4-year player development arc.
EuroLeague / Overseas	International scouting layer. Players can sign abroad, be scouted, and imported via draft or free agency.

3. Game Modes

There is one game mode: Career Mode. The player creates a coach and picks any team for their first job regardless of reputation. All subsequent jobs are governed by reputation, openings, and interviews — exactly like FM. Staying at a team requires performing well enough to keep the job. There is no single-team lock-in mode.

Manager Creation

Manager creation shapes the coach's starting reputation, coaching attributes, and tendencies. Inspired by FM26's creation flow, it consists of five steps.

Step 1 — Playing Career

What level did the coach reach as a player? This is the biggest single driver of starting reputation.

Option	Description	Rep Bonus
NBA Legend	Defined an era, won championships, highest individual honours	+40
NBA Veteran	Played significant minutes, respected career	+30
NBA Journeyman	Reached the league, bounced between rosters	+20
G League Pro	Career spent primarily in the G League	+12
College Star	Standout college player, never made the pros	+8

Overseas Pro	Played professionally abroad	+8
No Playing Career	Never played professionally	+0

Step 2 — Non-Playing Background

Experience outside of playing enhances the coaching profile. A recognition multiplier (Global ×1.5, National ×1.2, Regional ×1.0, Local ×0.7) is applied to the bonus.

Role	Sub-roles	Rep Bonus
Coaching	Head Coach, Assistant Coach, Player Development Coach, Skills Trainer	+5 to +15
Front Office	Scout, GM Assistant, Analytics Director	+5 to +10
Media & Broadcasting	Analyst, Commentator, Pundit	+3 to +8
Medical & Performance	Athletic Trainer, Strength & Conditioning Coach	+2 to +5
None	No non-playing background	+0

Step 3 — Coaching Qualifications

USA Basketball certification level. Affects coaching attributes and reputation — this is about formal credentials, not experience.

Qualification	Rep Bonus	Coaching Attr Impact
USA Basketball Gold License	+20	All coaching attrs Good–Very Good
USA Basketball Silver License	+12	Most attrs Average–Good
USA Basketball Bronze License	+6	Attrs Average
No Qualifications	+0	Attrs Poor–Average

Step 4 — Coaching Style (pick up to 3)

Defines the coaching philosophy and directly influences in-game attributes and team tendencies. Fewer choices = stronger focus. More choices = more versatile but less dominant in any area.

Style	Philosophy	Attribute Effect
Offensive Mastermind	Build systems that maximise scoring	+Offensive Systems, +Tactical Flexibility
Defensive Architect	Stopping the other team is everything	+Defensive Systems, +Tactical Flexibility
Player Developer	Unlocking potential is your gift	+Player Development, +Working With Youth
Pace & Space	Fast, spread the floor, three-point heavy	+Offensive Systems, tendency: High Tempo
System Builder	Rigid scheme, everyone fits the system	+Defensive Systems, +Tactical Flexibility
Players' Coach	Relationships drive everything	+People Management, +Motivating
Analyst	Data-driven decisions at every level	+Tactical Flexibility, +Scouting

Entertainer	Fans pay to watch your teams play	+Offensive Systems, tendency: High Risk/Reward

Step 5 — Personality (pick up to 2)

Shapes how the coach interacts with players, staff, and the media. Influences relationships, locker room dynamics, and certain reputation calculations.

Personality	Description	Attribute Effect
Driven	Relentless focus on winning	+Authority, +Determination
Disciplined	High standards, accountability above all	+Authority, tendency: Defensive
Players' Advocate	You fight for your players	+People Management, +Motivating
Resilient	Bounces back from adversity	+Determination, +Adaptability
Inspirational	Elevates everyone around you	+Motivating, +Authority
Accomplished	Thrives at any level in any situation	+Adaptability, reputation ×1.1

Coaching Attributes

The coach has two groups of attributes set by the creation choices above. These are expressed descriptively (Poor, Average, Good, Very Good, Outstanding) rather than as raw numbers.

Coaching Attributes	Mental Attributes
Offensive Systems	Adaptability
Defensive Systems	Authority
Player Development	Determination
Scouting	People Management
Motivation	Motivating
Tactical Flexibility	—
Fitness & Conditioning	—
Working With Youth	—

Starting Reputation Formula

```
reputation = playing_career_bonus            + (non_playing_bonus × recognition_multiplier)
+ qualification_bonus            × personality_multiplier  -- 1.1 if Accomplished, else 1.0
Reputation → Starting Job Tier:   0-20:   G League assistant or head coach at lower level   21-35:
G League head coach   36-50:  NBA assistant / first-time NBA head coach candidate   51-65:  Mid-
tier NBA head coach   66-80:  Top-tier NBA job candidate   81+:    Contender-level NBA teams
available from day one  Note: First job can be any team regardless of reputation. All subsequent
jobs are governed by reputation and openings.
```

Career Progression

Attribute	Effect

Reputation (0–100)	Gates job offers after first job. Below 35 = G League only. Above 65 = NBA mid-tier. Above 80 = contenders.
League Reputation	Separate scores per tier. NBA rep doesn't transfer directly to G League and vice versa.
Career Record	Wins, losses, playoff appearances, championships. Shown on coaching profile.
Contract	Current team, salary, years remaining. Fired = reputation penalty. Resign = small bonus.
Coaching Attributes	Can improve over time through coaching courses, experience, and staff development.

4. Technical Architecture

The project follows a strict separation of concerns: simulation logic never touches UI code, and all game state flows through global singletons (autoloads).

Layer Overview

Layer	Purpose
UI Layer (Godot scenes)	Reads from WorldState and emits signals via EventBus. No simulation logic.
Autoloads (global singletons)	EventBus, WorldState, Database, Clock, GameManager — always in memory.
Simulation Engine (GDScript)	Pure logic classes: MatchEngine, SeasonScheduler, DraftEngine, TradeEngine, PlayerDev.
Entity Layer (GDScript classes)	Data objects: Player, Team, Coach, Contract. Loaded into WorldState at game start.
Systems Layer	LeagueLoader, ScoutingSystem, NewsEngine — orchestration and cross-cutting concerns.
Data Layer (SQLite + JSON)	SQLite for saves and runtime data. JSON configs define league rules and structure.

Autoload Singletons

Singleton	Responsibility
EventBus.gd	Signal hub. All cross-system communication goes through here. No direct node references.
Database.gd	SQLite connection, schema creation, all query helpers. All DB access goes through this.
WorldState.gd	Master game state: all teams, all players, coach, standings, schedule, cap figures, TV deal tracking.
Clock.gd	Advances game calendar. Triggers match days, milestones, weekly standings rebuild.

GameManager.gd	New game setup (calls LeagueLoader), save/load orchestration, screen transitions.

File Structure

```
basketball-manager/   addons/godot-sqlite/              ← SQLite plugin   autoloads/
← 5 global singletons   core/entities/                    ← Player.gd, Team.gd
core/engine/match/                ← MatchEngine, PossessionSim   core/engine/season/
← SeasonScheduler, PlayoffEngine   core/engine/personnel/            ← DraftEngine, TradeEngine,
FreeAgency, ContractGenerator   core/engine/development/          ← PlayerGenerator, PlayerDev,
InjuryEngine   core/systems/                     ← LeagueLoader, ScoutingSystem, NewsEngine
data/leagues/nba/                 ← 5 JSON config files   data/leagues/g_league/            ←
g_league_teams.json   data/names/                       ← Procedural name banks   ui/screens/
← Hub, Squad, MatchDay, Draft, Scouting   ui/components/                    ← Reusable UI widgets
assets/fonts/
```

5. Player Attribute Model

Players have 53 attributes across 7 categories, rated 1–20 internally. Scouts see inaccurate versions based on scout_knowledge (0.0–1.0). Hidden attributes are never exposed directly — they surface as observable patterns in player behaviour and performance.

Attribute Categories

Category (count)	Attributes
Scoring (13)	Close Shot, Layup, Driving Dunk, Standing Dunk, Post Hook, Post Fade, Post Control, Mid-Range, Three-Point, Free Throw, Draw Foul, Hands, Shot IQ
Playmaking (6)	Ball Handling, Speed With Ball, Pass Accuracy, Pass Vision, Pass IQ, Off-Ball Movement
Defense (8)	Perimeter Defense, Interior Defense, Help Defense IQ, Pass Perception, Steal, Block, Defensive IQ, Defensive Consistency
Rebounding (2)	Offensive Rebounding, Defensive Rebounding
Athleticism (7)	Speed, Agility, Strength, Vertical, Stamina, Hustle, Overall Durability
Mental (9)	Anticipation, Composure, Concentration, Decision Making, Determination, Flair, Leadership, Teamwork, Work Ethic
Hidden (9)	Coachability, Consistency, Natural Fitness, Versatility, Adaptability, Injury Proneness, Important Matches, Concentration, Potential

Key Design Decisions

Decision	Detail
Scout visibility	44 visible attrs returned with noise ±6 at 0.0 knowledge, ±0 at 1.0. 9 hidden attrs never returned.
Delta-based potential	Each attribute has its own hidden growth ceiling. potential_deltas[attr] = how many more points it can grow.

Injury Proneness	High = bad (FM model). Not shown to scouts. Surfaces only through injury frequency patterns.
Important Matches	Hidden. High = rises to occasion. Low = nerves under pressure. Drives superstar trade demand system.
Flair	Visible. Drives improvised plays, high-risk/reward moves, fan attendance appeal.
OVR rating	Position-weighted display only. Simulation always uses individual attributes directly.

6. Player Generation System

PlayerGenerator.gd produces realistic players via a multi-layer pipeline. The same generator serves NBA rosters, G League rosters, and draft classes.

Generation Pipeline

Step	Detail
1. Roll tier	Weighted by league context. NBA: 20% Fringe / 40% Rotation / 28% Starter / 10% Star / 2% Superstar.
2. Roll archetype	Probability by tier: 20% Fringe → 100% Superstar. Archetype = primary skill + secondary skill.
3. Roll primary skill	Weighted by position. e.g. PG: 30% Passing, 25% Shot Creating, 20% Driving, 15% Three-Point, 10% Defending.
4. Roll secondary	Same pool as primary; primary skill gets 0.6× weight so same/same combos are allowed but not dominant.
5. Generate attributes	tier_mean + position_bias + primary_skill_bias + (secondary_bias × 0.6) + normal_noise(σ=2.0). Clamped 1–20.
6. Hidden attributes	Flat distribution: mean 8, σ=2.5. Fully independent of tier and position. Personality is random.
7. Age degradation	Physical attrs -0.6/year past age 26, capped -6. Mental attrs +0.3/year past 30, capped +3.
8. Potential deltas	Per-attribute growth ceilings. Younger + lower tier = more room to grow. Superstars near ceiling already.

Skill Groups & Available Skills by Position

Position	Available Primary Skills
PG	Passing & Ballhandling, Shot Creating, Three-Point, Driving & Finishing, Defending
SG	Passing & Ballhandling, Shot Creating, Three-Point, Driving & Finishing, Defending
SF	All 7 — includes Rebounding, Post Scoring, and Passing as primary options
PF	Three-Point, Driving & Finishing, Defending, Rebounding, Post Scoring

C	Three-Point, Driving & Finishing, Defending, Rebounding, Post Scoring

7. Salary Cap & Contract System

Full NBA salary cap fidelity. All cap figures stored as percentages of the current cap so they automatically scale each offseason when the cap increases.

Cap Structure

Rule	Detail
Starting cap	$136,000,000 (2025 season)
Luxury tax threshold	Cap × 1.10
First apron	Tax threshold × 1.053 — restrictions on MLE and trades
Second apron	Tax threshold × 1.105 — hard restrictions, pick limitations
Annual cap growth	6% base rate ± 2% variance each offseason
TV deal spike	Every 8–10 seasons: 20–30% cap jump. Telegraphed 2 seasons in advance via inbox.

Exceptions (all scale with cap)

Exception	Amount / Conditions
Full MLE	9% of cap / 4 years max. Teams over cap but under tax threshold.
Tax MLE	5% of cap / 3 years max. Teams over tax but under first apron.
BAE	3% of cap / 2 years max. Every other year. Unavailable above first apron.
Veteran min	1.8% (rookie), 2.8% (3–6 yrs), 3.8% (7+ yrs). All scale with cap.
Trade exception	Generated when outgoing salary > incoming. 1-year expiry. Non-combinable.

Contract Types

Type	Structure
Max contract	25%/30%/35% of cap based on experience. Up to 5 years.
Rookie scale (R1)	4 years, years 3–4 team options. Slot-based % of cap. 8% annual escalation.
Second round	2 years, year 2 team option. Veteran minimum range.
Two-way	$600K flat. Does not count against cap. 50 NBA day limit.
G League	$40K flat 1-year deal. No escalation. Resets each season.
Free agent deal	Bird rights determine max years and escalation rate (8% Bird, 5% non-Bird).

Luxury Tax Brackets

Amount Over Tax Threshold	Rate (per $1 over)
$0 – $5M	$1.50 (repeater: $2.50)
$5M – $10M	$1.75 (repeater: $2.75)
$10M – $15M	$2.50 (repeater: $3.50)
$15M – $20M	$3.25 (repeater: $4.25)
$20M – $25M	$3.75 (repeater: $4.75)
$25M+	$4.25 (repeater: $5.25)

Contract Staggering at World Gen

World gen uses a two-pass approach to ensure realistic payroll distribution. Pass 1 targets a spend level based on owner spending_aggressiveness (70–115% of cap). Pass 2 enforces cap compliance — trimming over-apron teams by converting non-star contracts to minimum deals, and bumping under-floor teams by raising fringe player salaries. Years remaining are staggered by tier and age so the entire league does not become free agents simultaneously.

8. Team & Owner System

Each Team object holds identity, owner personality, roster IDs, financials, standings, and draft pick ownership. Full G League Team objects exist with their own identity and affiliate links — G League teams inherit owner personality from their NBA parent.

Owner Attributes

Attribute	Effect
tax_willingness	0.0–1.0. How far above the luxury tax the owner will go. Ballmer = 1.0. Reinsdorf = 0.45.
spending_aggressiveness	0.0–1.0. How quickly they spend to cap vs staying conservative.
patience	0.0–1.0. Affects firing threshold (base 8 + patience×8 = range 8–16 pressure points).
ambition	0.0–1.0. Pushes expectation tier up (≥0.8) or down (≤0.39) from engine baseline.
will_tank	Boolean. Owner immune to pressure during explicit tank seasons.

Draft Pick Protection

Picks are tracked per-team with full protection schedules. Pick ID format: TEAM_YEAR_R1 (e.g. BOS_2027_R1). Each pick stores a protection_schedule array with per-year protection rules. Protection types: top_n, range,

lottery, unprotected. When protection triggers, the pick rolls forward, converts to a second rounder, reverts to original team, or becomes unprotected — based on the if_triggers and fallback fields.

9. Expectations & Coaching Pressure

Each preseason, the engine sets an expectation tier based on roster strength, adjusted by owner ambition. At season end, actual results are scored and compared against expectations. Pressure accumulates and fires the coach_fired event when it exceeds the owner's firing threshold.

Expectation Tiers

Tier	Target Score	Description
title_contender	5–8	Win the championship
deep_run	4–6	Conference finals or better
playoff_team	2–4	Top 6 seed in conference
play_in	0–2	Qualify for play-in (seeds 7–10)
development	-1–1	Win 30–40 games, develop youth
tank	-2–0	Lottery positioning

Performance Scoring

Result	Points
Championship	+5
Finals appearance	+3
Conference finals	+2
Second round exit	+1
First round exit	0
Play-in eliminated	-1
Missed play-in	-3
60+ wins	+3 | 50–59 wins: +2 | 40–49: +1 | 30–39: 0 | 20–29: -1 | <20: -2

Superstar Trade Demand System

Players with Important Matches ≥ 14 (hidden) have an active title timeline. After 2 consecutive missed expectation seasons (or 1 if Important Matches ≥ 17), demand escalates through three levels: unhappy (front office meeting), trade_request (formal request, morale penalty if unresolved), and demand (public demand, locker room morale hit on all teammates).

10. Match Simulation Engine

The match engine uses a possession-based simulation model. Each possession generates a sequence of events (pass, drive, screen, shot) and resolves outcomes using relevant player attributes weighted against opponent attributes and contextual factors.

Key Design Decisions

Decision	Detail
Score distribution	Normal distribution (not Poisson) — basketball scores are high and clustered
Possessions per game	~100 per game; team pace rating affects this
Fatigue model	Stamina degrades attributes throughout the game; rotation management matters
Clutch factor	Composure + Important Matches activate in final 2 minutes of close games (margin ≤5)
Home court advantage	~3.5% boost to home team performance
Simulation fidelity	Full engine for player's games; lightweight result generator for background games

11. League Loader & World Generation

LeagueLoader.gd is a one-shot system called by GameManager.new_game(). It reads all JSON configs, constructs all Team and Player objects, generates staggered contracts, enforces cap compliance, and saves everything to the DB in batch transactions.

World Gen Sequence

Step	Action
1	Load all JSON configs into WorldState.league_configs
2	Set per-league salary caps from config
3	Initialise TV spike schedule (random year within 8–10 season window)
4	Build 30 NBA Team objects from nba_teams.json
5	Build 30 G League Team objects from g_league_teams.json
6	Assign 5 years of draft picks (both rounds) to all 30 NBA teams
7	Generate NBA rosters: 15 players × 30 teams = 450 players
8	Generate G League rosters: 10 players × 30 teams = 300 players
9	Generate draft class: 60 prospects for next draft

10	Generate staggered contracts for all ~810 players
11	Two-pass cap compliance: trim over-apron, bump under-floor
12	Batch save all players, teams, and contracts to DB
13	Start player_tenure records for all rostered players
14	Rebuild initial standings (all 0–0)

12. Development Roadmap

Feature	Description	Status
Godot 4 project setup	Project structure, folders, settings	**Done**
SQLite plugin	godot-sqlite via AssetLib, schema created	**Done**
Autoload singletons	EventBus, Database, WorldState, Clock, GameManager	**Done**
Player entity	53 attributes across 7 categories, scout visibility, delta potential	**Done**
PlayerGenerator	Tier system, archetype system, position biases, age degradation	**Done**
Team entity	Cap helpers, roster management, pick protections, standings, pressure system	**Done**
Database layer	Full schema: 53 attrs, player_tenure, contracts, teams, picks	**Done**
League JSON configs	5 NBA files + g_league_teams.json — cap, calendar, teams, expectations	**Done**
ContractGenerator	World gen, free agency, draft night, G League — staggered + bird rights	**Done**
LeagueLoader	One-shot world gen: 810 players, 60 teams, contracts, tenure, standings	**Done**
WorldState updates	Cap system, TV spike tracking, standings rebuild, typed accessors	**Done**
Match engine (core)	Possession sim, shot resolution, box score output	**Planned**
Season scheduler	82-game schedule, standings, tiebreakers	**Planned**

Draft engine	Prospect generation, scouting, lottery, draft night UI	**Planned**
Trade engine	AI evaluation, salary matching, pick trades, trade UI	**Planned**
Free agency system	Player decisions, bidding, two-way contracts	**Planned**
Player development	Aging curves, training, potential reveal	**Planned**
Inbox / news engine	FM-style message inbox: contract offers, scout reports, board expectations	**Planned**
Hub UI screen	Main dashboard, inbox, calendar, quick stats	**Planned**
Squad / tactics screen	Roster management, lineup setting, rotations	**Planned**
Match day screen	Text play-by-play, live score, substitutions	**Planned**
Career mode logic	Job offers, reputation system, coach progression, owner expectations	**Planned**
Scouting system	Scout ratings, report accuracy, overseas scouting	**Planned**
Save / load system	Full game state serialization via SQLite	**Planned**
G League season loop	50-game season, own standings, promotion/call-up pipeline	**Planned**
College pipeline	Recruiting, 4-year development, draft entry	**Planned**
Overseas leagues	EuroLeague config, international transfers	**Planned**

13. Key References

Reference	What We Take From It
Football Manager	Attribute simulation, reputation system, text match engine, scouting inaccuracy, inbox system
OOTP Baseball	Current + potential dual ratings, financial coefficient, league config flexibility
NBA 2K (attribute model)	53-attribute model adapted from 2K's attribute categories and archetype system
OpenFootManager (GitHub)	Open source manager: Rust sim engine + React UI. Architecture separation pattern.

open-source-ball (GitHub)	Swift iOS college hoops sim. College pipeline logic, chunked save approach.
xg_simulator.py (Arya)	Existing Poisson/Monte Carlo match sim. Informs possession engine design philosophy.

14. Design Notes & Decisions

Licensing

All teams and players ship with real NBA names for development. Architecture supports community database mods to swap names if needed. Fictional names are a deferred decision.

No 3D Engine (by design)

The 3D match view is explicitly out of scope for v1. Text-based simulation with rich play-by-play commentary is the target. A 2D match view may be added later as a non-blocking feature.

Simulation Fidelity Tiers

Following Football Manager's approach: a full possession-by-possession engine for the player's team's games, and a lightweight result generator for background games. This keeps season simulation fast while maintaining depth where the player is watching.

GDScript vs C#

GDScript chosen for Python-like syntax matching existing background and tight Godot integration. If performance becomes an issue, the simulation engine can be ported to C# without touching UI code due to clean architecture separation.

Known Issue — World Gen Roster IDs

Roster IDs on Team objects are populated during player generation before DB save assigns real auto-incremented IDs. Requires a post-save ID fixup pass. To be resolved during first Godot integration test.